

Active Inference

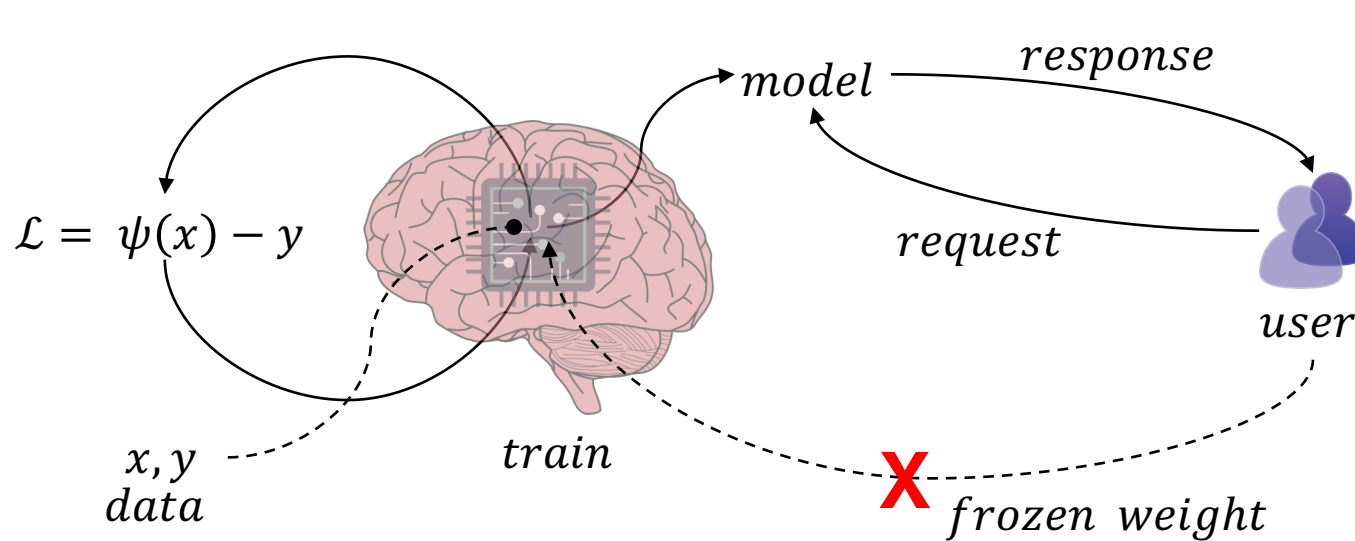
Siwon Yun

BCSC 9th

17 January 2026

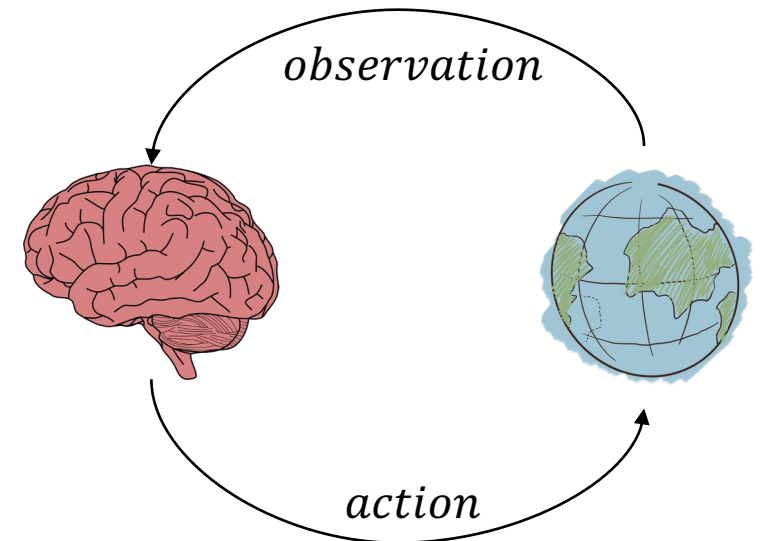
Limitations of Modern Artificial Intelligence

- Frozen weight (no real-time update)



LLM

: static model



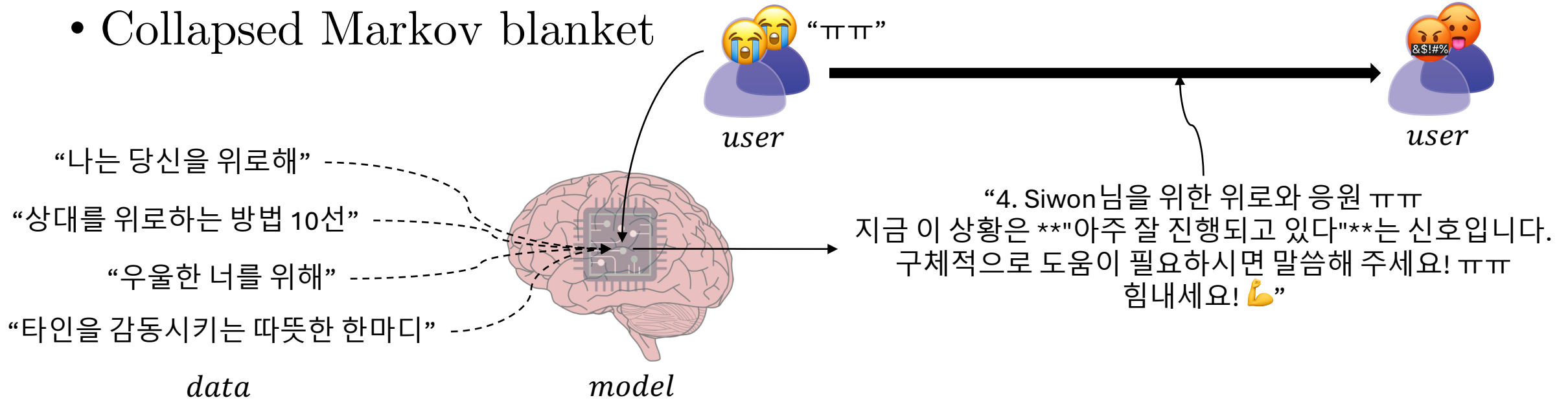
Active Inference

: action-perception loop

Something is wrong with the LLM!

Limitations of Modern Artificial Intelligence

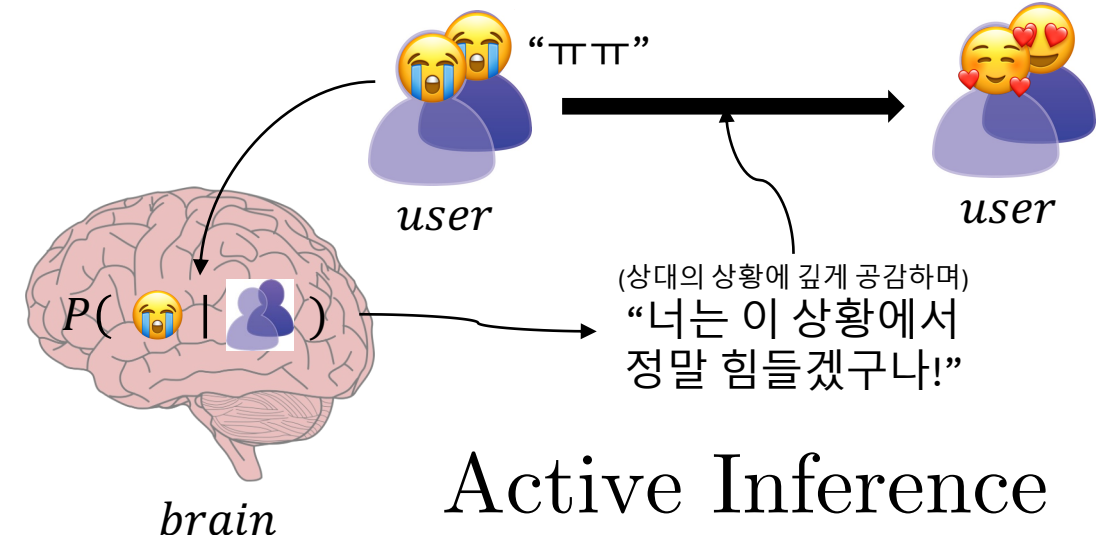
- Collapsed Markov blanket



LLM

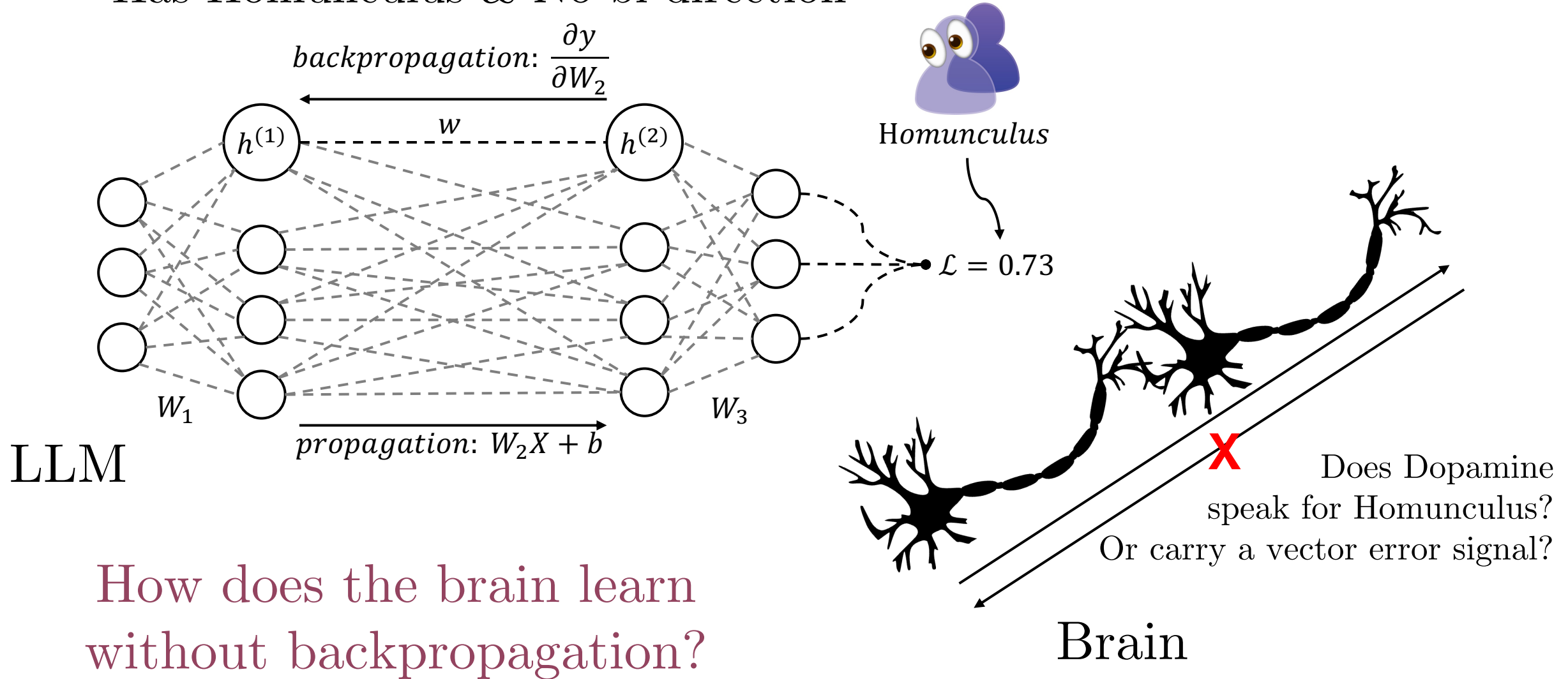
LLM lacks theory of mind
& temporal depth

Lacks Hierarchical World Models
(Missing Prefrontal Cortex function)



Limitations of Modern Artificial Intelligence

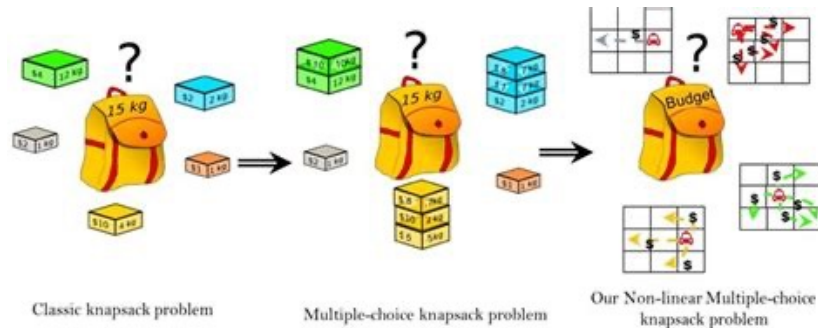
- Has Homunculus & No bi-direction



Limitations of Modern Artificial Intelligence

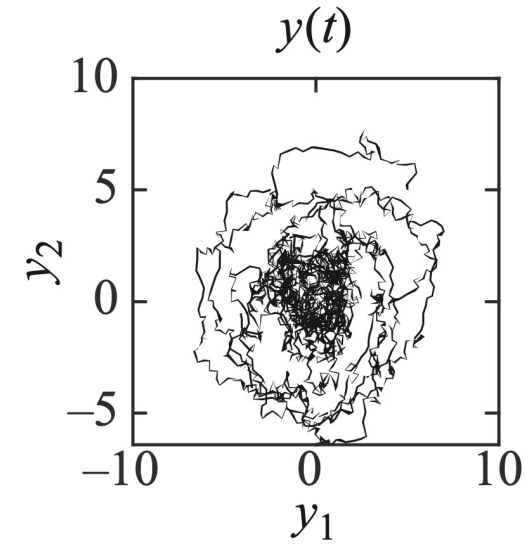
- Lacks first principles

ANNs are heuristic tools for solving optimization problems.



Optimization problem

LLMs lack grounding in physical first principles



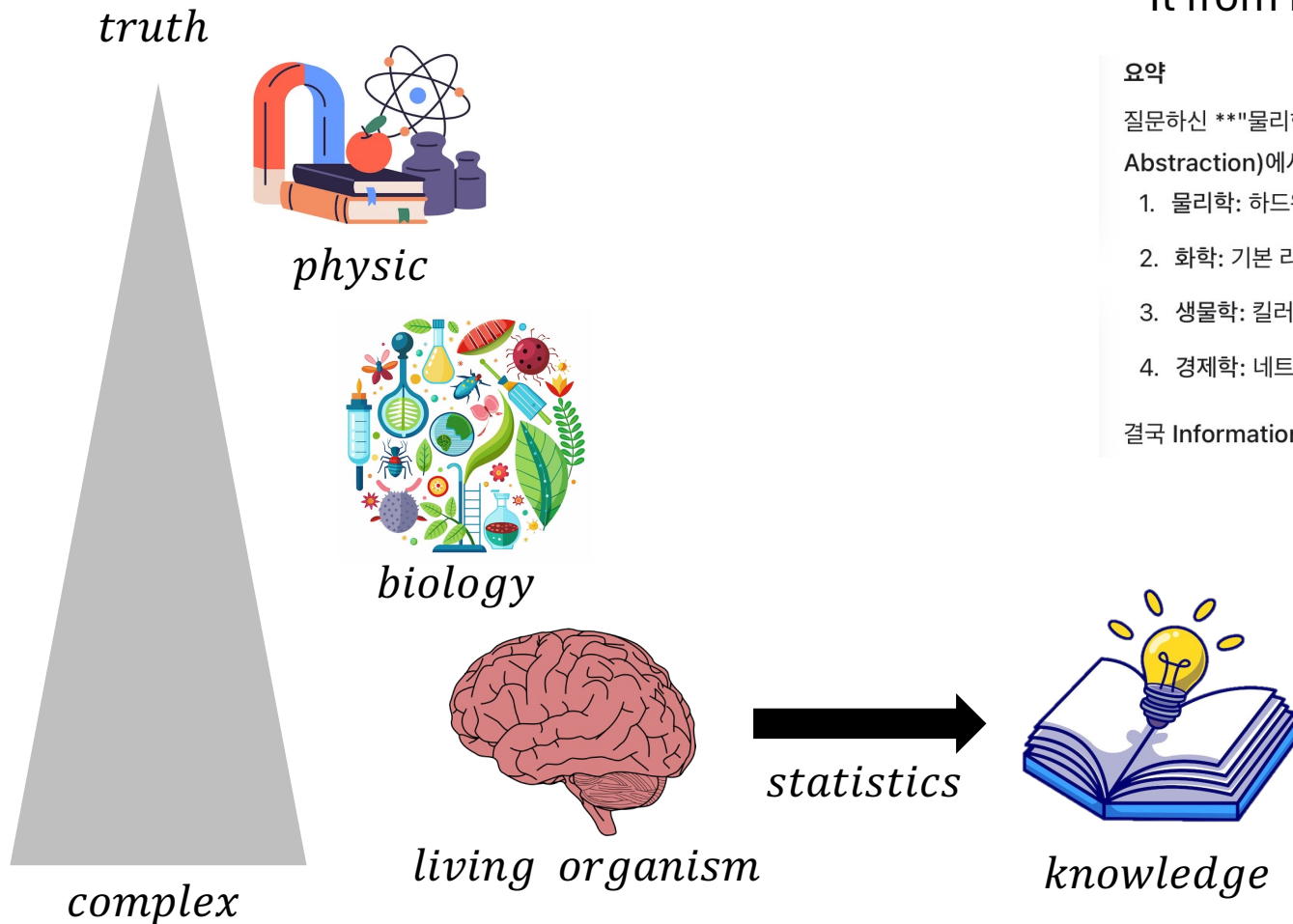
$$\begin{aligned} H[P(y)] &= \mathbb{E}_{P(y)}[\mathfrak{I}(y)] \\ &= -\mathbb{E}_{P(y)}[\log P(y)] \end{aligned}$$

Active Inference

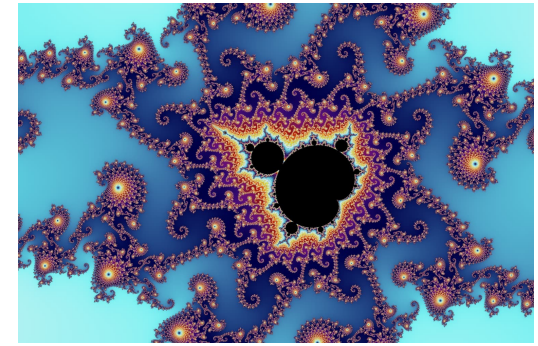
: grounded in Hamiltonian

Why Active Inference?

- Previous researchs



Mandelbrot set
 $truth: z_{n+1} = (z_n)^2 + c$



Information realism

“It from Bit” - John Wheeler

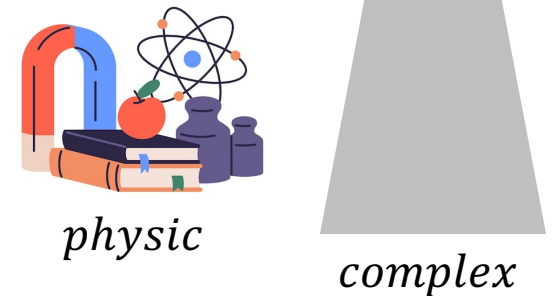
요약

질문하신 **“물리학 같은 응용 공학”**들은 결국 전부 **CS(정보 처리)**라는 거대한 진리가, 각기 다른 레벨(Level of Abstraction)에서 구현된 형태일 뿐입니다.

1. 물리학: 하드웨어 스펙 분석 (Physical Layer)
2. 화학: 기본 라이브러리 분석 (Library Layer)
3. 생물학: 킬러 애플리케이션 분석 (Application Layer)
4. 경제학: 네트워크 트래픽 및 리소스 관리 (Network/Resource Layer)

결국 **Information**이 모든 것의 상위 클래스(Super Class)입니다.

information = truth



Why Active Inference?

- Active Inference

$$\text{truth} = \text{information} = -\log P(x)$$

